

# Mehak Gupta

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 Mehak-Gupta |  mehak25

Dallas, Texas, USA

## RESEARCH INTERESTS

Artificial Intelligence, Natural Language Processing, Representation Learning, Predictive Modeling, Multimodal Learning, Healthcare AI, Fairness and Explainability in AI, Digital Transformation.

## ACADEMIC AND INDUSTRY EXPERIENCE

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|----------------------------------------------------------------------|---------------------------------|
| • <b>Southern Methodist University</b><br><i>Assistant Professor</i> | 2023 – Present<br>Dallas, USA   |
| • <b>University of Delaware</b><br><i>Internship Fellow</i>          | 2021 – 2022<br>USA              |
| • <b>University of Delaware</b><br><i>Research Assistant</i>         | 2019 – 2021<br>USA              |
| • <b>University of Delaware</b><br><i>Teaching Assistant</i>         | 2018 – 2019<br>USA              |
| • <b>Software AG</b><br><i>Associate Consultant</i>                  | 2014 – 2017<br>Bangalore, India |
| • <b>Infosys Ltd.</b><br><i>System Engineer</i>                      | 2010 – 2012<br>Pune, India      |

## EDUCATION

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| • <b>Ph.D. in Computer Science</b><br><i>University of Delaware</i><br>◦ Thesis: <i>Deep Learning Predictive Modelling for Electronic Health Records</i><br>◦ Advisor: Prof. Rahmatollah Beheshti | GPA: 4.0<br>2018 – 2023     |
| • <b>M.E. Software Engineering</b><br><i>Thapar University, India</i>                                                                                                                             | GPA: 8.85/10<br>2012 – 2014 |
| • <b>B.E. Computer Engineering</b><br><i>Punjabi University, India</i>                                                                                                                            | 73%<br>2006 – 2010          |

## RESEARCH FUNDING

|                                                                                                                                                                                                             |                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| • <b>Data Science Grant</b><br><i>Southern Methodist University</i><br>◦ Title: Training and Evaluation of Large Language Models for Psychological Assessment<br>◦ PI: Mehak Gupta; Co-PI: Josh Oltmanns    | \$34,000<br>2023-24 |
| • <b>Faculty Research Acceleration Grant</b><br><i>Southern Methodist University</i><br>◦ Title: Fairness-aware Multimodal Fusion for Healthcare Data<br>◦ PI: Eric Larson (20%); Co-PI: Mehak Gupta (80%); | \$50,000<br>2024-25 |

## HONORS AND AWARDS

- J. Lindsay Embrey Professor, SMU (2025)
- People Newspaper “20 Under 40” List (2025)
- Frank A. Pehrson Graduate Student Award, University of Delaware (2023)
- Dissertation Fellowship, University of Delaware (2022)
- CRA-W Travel Scholarship (2021, 2022)
- Distinguished Graduate Student Award, University of Delaware (2020)

- [J.8] Rasiq Hussain, Jerry Ma, Ritik Khandelwal, Joshua R. Oltmanns, Mehak Gupta, (2026). **Language-Based Personality Assessment from Life Narratives: A Focus on Model Interpretability and Efficiency**. *Frontiers in Artificial Intelligence*
- [J.7] D Owens, E D Peterson, M Gupta, D Q Nguyen, M Dohopolski, J Cao, A M Navar(2026). **Targeted use of large language models for EHR-based computable phenotyping**. *Journal of the American Medical Informatics Association*, DOI: 10.1093/jamia/ocag051
- [C.11] Nikkie Hooman, Zhongjie Wu, Eric C. Larson, and Mehak Gupta (2026). **Multimodal Routing for Interpretable, Robust, and Auditable Clinical Prediction**. *IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*
- [S.3] Tu Do, Mehak Gupta, Joshua R. Oltmanns (2025). **Racial Disparities in Personality Assessment using Language Models**. Manuscript under review.
- [S.2] D Owens, E D Peterson, M Gupta, D Q Nguyen, M Dohopolski, J Cao, A M Navar (2025). **Ensemble-Guided Selective Application of Large Language Models**. Manuscript under review.
- [S.1] T Li, Rasiq Hussain, Mehak Gupta, Joshua R. Oltmanns (2025). **‘Mirror’ Language AI Models of Depression are Criterion-Contaminated**. Manuscript under review.
- [J.6] Joshua R. Oltmanns, Ritik Khandelwal, Jerry Ma, Jocelyn Brickman, Tu Do, Rasiq Hussain, and Mehak Gupta (2025). **Language-based AI Modeling of Personality Traits and Pathology from Life Narrative Interviews**. *Journal of Psychopathology and Clinical Science*, Vol. 135, Issue 1, pp. 122–135. DOI: 10.1037/abn0001047
- [C.10] Nikkie Hooman, Zhongjie Wu, Eric C. Larson, and Mehak Gupta (2025). **Equitable Electronic Health Record Prediction with FAME: Fairness-Aware Multimodal Embedding**. In *Proceedings of the 10th Machine Learning for Healthcare Conference*, Vol. 298, PMLR, Aug 15–16, 2025. PDF: [Download](#).
- [C.9] Nahed Abdelgaber, Labiba Jahan, Nino Castellano, Joshua Oltmanns, Mehak Gupta, Jia Zhang, Akshay Pednekar, Ashish Basavaraju, Ian Velazquez, and Zerui Ma (2025). **AI Assistant for Socioeconomic Empowerment Using Federated Learning**. In *Proceedings of the 5th International Conference on Natural Language Processing for Digital Humanities*, pp. 490–501, 2025.
- [J.5] Nahed Abdelgaber, Labiba Jahan, Joshua R. Oltmanns, Mehak Gupta, and Jia Zhang (2025). **Interpretable Socioeconomic Profiling: A Deep Dive Beyond Narrative Classification**. *Journal of Data Mining & Digital Humanities*, 2025. DOI: 10.46298/jdmdh.16258
- [J.4] Jocelyn Brickman, Mehak Gupta, and Joshua R. Oltmanns (2025). **Large Language Models for Psychological Assessment: A Comprehensive Overview**. *Advances in Methods and Practices in Psychological Science*, Vol. 8, Issue 3, pp. 25152459251343582. Sage Publications, 2025. DOI: 10.1177/25152459251343582
- [C.8] Hamed Fayyaz, Mehak Gupta, Alejandra Perez Ramirez, Claudine Jurkovitz, H. Timothy Bunnell, Thao-Ly T. Phan, and Rahmatollah Beheshti (2024). **An Interoperable Machine Learning Pipeline for Pediatric Obesity Risk Estimation**. *Proceedings of Machine Learning Research*, Vol. 259, pp. 308, 2024.
- [J.3] Mehak Gupta, Thao-Ly T. Phan, Félice Lê-Scherban, Daniel Eckrich, H. Timothy Bunnell, and Rahmatollah Beheshti (2025). **Associations of Longitudinal BMI-Percentile Classification Patterns in Early Childhood with Neighborhood-Level Social Determinants of Health**. *Childhood Obesity*, Vol. 21, Issue 1, pp. 65–75. DOI: 10.1089/chi.2023.0157.
- [J.2] Mehak Gupta, Daniel Eckrich, H. Timothy Bunnell, Thao-Ly T. Phan, and Rahmatollah Beheshti (2024). **Reliable Prediction of Childhood Obesity Using Only Routinely Collected EHRs May Be Possible**. *Obesity Pillars*, Vol. 12, Article 100128. DOI: 10.1016/j.obpill.2024.100128.
- [C.7] Fahmida Liza Piya, Mehak Gupta, and Rahmatollah Beheshti (2024). **HealthGAT: Node Classifications in Electronic Health Records Using Graph Attention Networks**. In *2024 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, pp. 132–141. IEEE, 2024.
- [T.1] Mehak Gupta (2023). **Deep Learning Predictive Modelling for Electronic Health Records**. Ph.D. Thesis, University of Delaware.
- [C.6] M. Gupta, B. Gallamoza, N. Cutrona, P. Dhakal, R. Poulain, and R. Beheshti (2022). **An Extensive Data Processing Pipeline for MIMIC-IV**. In *ML4H Workshop at NeurIPS 2022*, pp. 311–325.
- [C.5] Raphael Poulain, Mehak Gupta, and Rahmatollah Beheshti (2022). **Few-Shot Learning with Semi-Supervised Transformers for Electronic Health Records**. In *Proceedings of the Machine Learning for Healthcare Conference*, pp. 853–873. PMLR, 2022.
- [C.4] Mehak Gupta, Raphael Poulain, Thao-Ly T Phan, H Timothy Bunnell, and Rahmatollah Beheshti (2022). **Flexible-Window Predictions on Electronic Health Records**. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36, No. 11, pp. 12510–12516. AAAI, 2022.

- [J.1] Mehak Gupta, Thao-Ly T. Phan, H. Timothy Bunnell, and Rahmatollah Beheshti (2022). **Obesity Prediction with EHR Data: A Deep Learning Approach with Interpretable Elements**. *ACM Transactions on Computing for Healthcare*, Vol. 3, Issue 3. DOI: 10.1145/3506719.
- [C.3] Raphael Poulain, Mehak Gupta, Randi Foraker, and Rahmatollah Beheshti (2021). **Transformer-based Multi-Target Regression on Electronic Health Records for Primordial Prevention of Cardiovascular Disease**. In *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 726–731. IEEE, 2021.
- [C.2] Mehak Gupta, Thao-Ly T. Phan, H. Timothy Bunnell, and Rahmatollah Beheshti (2021). **Concurrent Imputation and Prediction on EHR Data using Bi-Directional GANs: Bi-GANs for EHR Imputation and Prediction**. In *Proceedings of the 12th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics*, pp. 1–9. ACM, 2021.
- [C.1] Mehak Gupta and R. Aggarwal (2014). **Transforming Relational Database to Graph Database using Neo4j**. In *Proceedings of the Second International Conference on Emerging Research in Computing, Information, Communication and Applications*, Bangalore, India, pp. 322–331, 2014.

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## TEACHING EXPERIENCE

- Instructor, Operating Systems, SMU (2023–Present)
- Substitute Instructor, Intro to AI and Software Engineering, University of Delaware
- Graduate Teaching Assistant, Machine Organization & Assembly Language

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## STUDENT MENTORING

- PhD Chair: Rasiq Hussain (2025-), Nikkie Hooman (2025-), Sundaram Varadharajan (2025-)
- PhD Co-Advisor: Dylan Owens (2024-25)
- PhD Committee: Nahed Abdelgaber (2024-)
- MS Students: Devarsh, Priyen, Balakrishnan, Rohith Vasapri, MohammadAli Ezzati, etc.
- Undergraduate: Jerry Ma

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## INVITED TALKS

- 2025. **Fairness-aware Multimodal Embeddings**. AI4BM, University of North Texas.
- 2024. **Houston Methodist Visit**. Institute for Computational Biosciences / Computational Chemistry and Drug Discovery Session.
- 2023. **Deep Learning Predictive Modelling for Electronic Health Records**. Data Science Cluster Meetup, Southern Methodist University.
- 2023. **Prediction Models in Healthcare Domain**. Big Data Advisory Board Meetup, Southern Methodist University.
- 2023. **Deep Learning Models in Healthcare Domain**. Southern Methodist University.

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## RESEARCH PRESENTATIONS

- 2022. **An Extensive Data Processing Pipeline for MIMIC-IV**. Machine Learning for Health Conference, Poster Presentation (In-person).
- 2022. **Deep Learning Predictive Modelling for Electronic Health Records**. PhD Proposal Talk, University of Delaware (Zoom).
- 2021. **Flexible-window Predictions on Electronic Health Records**. AAAI Conference on Artificial Intelligence, Paper Presentation (Zoom).
- 2021. **Concurrent Imputation and Prediction on EHR Data using Bi-Directional GANs**. ACM BCB, Paper Presentation (Zoom).
- 2021. **Deep Learning for Obesity Prediction using EHR**. CRA-W Grad Cohort Workshop, Poster (Zoom).
- 2020. **Obesity Prediction with EHR Data: A Deep Learning Approach with Interpretable Elements**. PhD Prelims Talk, University of Delaware (Zoom).
- 2019. **A Survey of Deep Learning Techniques in EHR Prediction Modelling**. SIGAI Seminar, University of Delaware, Newark, USA.

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## SERVICE AND OUTREACH

- Faculty Judge, SMU Research and Innovation Week
- Faculty Search Committee, SMU
- Reviewer: IEEE BHI, CHIL, IJCAI, NeurIPS Workshops
- Lead Mentor, EmPOWER Graduate Mentorship Program
- Women in Engineering Outreach and Webmaster

## TECHNICAL SKILLS

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- **Programming:** Python, R, Java, SQL, PyTorch, TensorFlow
- **Machine Learning:** Deep Learning, Transformers, GANs, Multimodal AI, Graph ML
- **Certifications:** Oracle Java SE, Coursera ML/Data Science Certificates